

Appendix: Practical Measurement and Control of Quality Factor in Piano Strings

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This appendix provides practical guidelines for measuring and controlling the quality factor Q in piano strings, as discussed in the main paper. The MATLAB codes presented here allow the reader to reproduce the simulations and gain insight into the role of Q in determining the attack time and spectral friction parameter κ .

1 Practical Definitions of Q

From an engineering perspective, Q can be defined and measured in three equivalent ways:

1. **Energy ratio:**

$$Q = 2\pi \times \frac{\text{Energy stored}}{\text{Energy lost per cycle}}.$$

2. **Resonance bandwidth:**

$$Q = \frac{f_0}{\Delta f},$$

where Δf is the half-power bandwidth.

3. **Decay rate:**

$$Q = \frac{\pi}{\delta}, \quad \delta = \ln \left(\frac{A_1}{A_2} \right),$$

with δ being the logarithmic decrement between successive peaks.

2 MATLAB Code for Practical Q Implementation

The following MATLAB code simulates the effect of Q on the string response, demonstrates measurement techniques, and shows how to design a piano for desired Q values.

2.1 Effect of Q on String Decay Time

The code below generates damped sinusoidal signals for different Q values (600, 1500, 3000) and plots the time-domain response. It also computes the T_{60} decay time.

```
1 % Practical Q implementation for piano string design
2 clear; close all; clc;
3
4 %% Method 1: Measurement from impulse response
5 fs = 10000;           % sampling frequency
6 t = 0:1/fs:2;         % 2 seconds of signal
```

```

7 f0 = 261.6;          % C4 frequency
8
9 % Different Q values for the same note
10 Q_values = [600, 1500, 3000]; % from your paper's Table 2
11 colors = {'r', 'b', 'g'};
12
13 figure('Position', [100, 100, 1200, 400]);
14
15 for i = 1:length(Q_values)
16     Q = Q_values(i);
17     zeta = 1/(2*Q); % damping ratio
18
19     % Damped sinusoidal response
20     omega0 = 2*pi*f0;
21     omega_d = omega0 * sqrt(1 - zeta^2);
22     envelope = exp(-zeta * omega0 * t);
23     signal = envelope .* sin(omega_d * t);
24
25     % Plot
26     subplot(1,3,i);
27     plot(t, signal, colors{i}, 'LineWidth', 1.5);
28     hold on;
29     plot(t, envelope, 'k--', 'LineWidth', 1);
30     plot(t, -envelope, 'k--', 'LineWidth', 1);
31     xlim([0 0.5]);
32     xlabel('Time (s)');
33     ylabel('Amplitude');
34     title(sprintf('Q = %d', Q));
35     grid on;
36
37     % Measure decay time to -60dB
38     decay_60dB = -log(0.001) / (zeta * omega0); % time to -60dB
39     text(0.3, 0.8, sprintf('T_{60} = %.2f s', decay_60dB), ...
40          'FontSize', 10, 'BackgroundColor', 'w');
41 end
42 sgtitle('Effect of Q on String Decay Time');

```

Listing 1: Effect of Q on decay time

2.2 Factors Affecting Q in Piano Strings

The next code creates a pie chart of the main contributors to Q and shows practical methods for adjusting Q .

```

1 %% Method 2: Practical Q control in instrument making
2 figure('Position', [100, 100, 1000, 600]);
3
4 % Factors affecting Q in piano strings
5 factors = {'Wire material (thermal conductivity)', 'Support stiffness', 'Air
6           viscosity', ...
7           'Internal friction', 'Sound radiation', 'Soundboard coupling'};
8
9 impact = [25, 30, 15, 20, 5, 5]; % approximate percentage
10
11 % Plot pie chart
12 subplot(2,2,[1,2]);
13 pie(impact, factors);
14 title('Factors Affecting Q in Piano Strings');
15
16 % Practical methods for adjusting Q

```

```

15 subplot(2,2,3);
16 methods = {'Change wire material', 'Change tension', 'Modify supports', 'External
    damping'};
17 Q_range = [500 2000; 1000 3000; 800 2500; 100 800]; % achievable ranges
18 bar(Q_range);
19 set(gca, 'XTickLabel', methods, 'XTickLabelRotation', 45);
20 ylabel('Achievable Q range');
21 title('Practical Methods to Adjust Q');
22 grid on;
23
24 subplot(2,2,4);
25 % Empirical relation between Q and physical parameters
26 L_lambda = 0.4:0.05:0.6; % typical L/ range
27 Q_needed = 183.714 ./ (0.688 + 8.801*L_lambda - 3); % condition >3
28 plot(L_lambda, Q_needed, 'b-', 'LineWidth', 2);
29 xlabel('L/ ');
30 ylabel('Minimum required Q');
31 title('Condition >0: Q > 183.714/(0.688 + 8.801(L/ ))');
32 grid on;

```

Listing 2: Factors influencing Q and adjustment methods

2.3 Laboratory Measurement of Q

This section simulates a realistic measurement with noise and estimates Q from peak amplitudes.

```

1 %% Method 3: Laboratory measurement of Q
2 figure('Position', [100, 100, 1000, 400]);
3
4 % Simulate a real measurement
5 t_meas = 0:1/fs:1;
6 Q_actual = 1500;
7 zeta_actual = 1/(2*Q_actual);
8 signal_meas = exp(-zeta_actual * omega0 * t_meas) .* sin(omega0 * t_meas);
9
10 % Add measurement noise
11 signal_meas = signal_meas + 0.01 * randn(size(signal_meas));
12
13 % Find peaks
14 [pks, locs] = findpeaks(signal_meas(1:5000), 'MinPeakHeight', 0.1);
15 t_peaks = t_meas(locs);
16
17 % Estimate Q from decay
18 if length(pks) >= 2
19     decay_rate = -log(pks(2)/pks(1)) / (t_peaks(2) - t_peaks(1));
20     Q_measured = pi / (decay_rate / f0);
21 end
22
23 % Plot signal and peaks
24 subplot(1,2,1);
25 plot(t_meas, signal_meas, 'b-');
26 hold on;
27 plot(t_peaks, pks, 'ro', 'MarkerSize', 8, 'MarkerFaceColor', 'r');
28 xlabel('Time (s)');
29 ylabel('Amplitude');
30 title('Impulse Response and Detected Peaks');
31 legend('Signal', 'Peaks');
32 xlim([0 0.5]);
33 grid on;

```

```

34
35 subplot(1,2,2);
36 semilogy(t_peaks, pks, 'bo-', 'LineWidth', 2);
37 xlabel('Time (s)');
38 ylabel('Peak Amplitude (log scale)');
39 title(sprintf('Measured Q = %.0f (actual = %.0f)', Q_measured, Q_actual));
40 grid on;

```

Listing 3: Laboratory measurement of Q

2.4 Active Control of Q

An advanced method using feedback to modify the effective Q is demonstrated below.

```

1 %% Method 4: Active Q control (advanced)
2 figure('Position', [100, 100, 800, 400]);
3
4 % Simulation of active damping with feedback
5 t_active = 0:1/fs:0.5;
6 Q_passive = 1500; % intrinsic Q
7 Q_target = 800; % desired Q for timbre change
8
9 zeta_passive = 1/(2*Q_passive);
10 zeta_target = 1/(2*Q_target);
11 zeta_active = zeta_target - zeta_passive; % additional damping needed
12
13 % Passive and active signals
14 signal_passive = exp(-zeta_passive * omega0 * t_active) .* sin(omega0 * t_active);
15 signal_active = exp(-zeta_target * omega0 * t_active) .* sin(omega0 * t_active);
16
17 plot(t_active, signal_passive, 'b-', 'LineWidth', 1.5);
18 hold on;
19 plot(t_active, signal_active, 'r-', 'LineWidth', 1.5);
20 xlabel('Time (s)');
21 ylabel('Amplitude');
22 legend('Intrinsic Q = 1500', 'Controlled Q = 800');
23 title('Active Q Control Using Feedback');
24 grid on;

```

Listing 4: Active Q control via feedback

2.5 Practical Guide for Piano Makers

Finally, a practical guide linking the paper's regression formula to real-world design choices.

```

1 %% Practical Guide for Piano Makers
2 figure('Position', [100, 100, 1000, 600]);
3
4 % Target Q for different registers
5 notes = {'C2 (bass)', 'C4 (mid)', 'C7 (treble)'};
6 desired_kappa = [3.4, 4.1, 5.9]; % from your paper
7 L_lambda_vals = [0.5, 0.45, 0.6]; % approximate L/ for each note
8 Q_needed = 183.714 ./ (0.688 + 8.801*L_lambda_vals - desired_kappa);
9
10 subplot(2,2,1);
11 bar(Q_needed);
12 set(gca, 'XTickLabel', notes);
13 ylabel('Required Q');

```

```

14 title('Target Q for Each Note');
15 grid on;
16
17 % Practical methods to achieve target Q
18 methods_table = {
19     'Low (C2)', 'Thicker wire + softer hammer felt', 'High Q (~3000)';
20     'Mid (C4)', 'Standard wire + wooden support', 'Medium Q (~1500)';
21     'High (C7)', 'Thin wire + hard support', 'Low Q (~600)'
22 };
23
24 subplot(2,2,[2,4]);
25 uitable('Data', methods_table, ...
26     'ColumnName', {'Note Range', 'Construction Method', 'Resulting Q'}, ...
27     'Position', [400, 100, 350, 150]);
28
29 % Summary formula
30 text(0.1, 0.5, ['Key relation:  $\kappa = 0.688 + 8.801(L/\lambda) - 183.714(1/Q)$ '
31     , ...
32     '\newline\nTo adjust  $\kappa$  (attack time):', ...
33     '\newline If  $\kappa$  is too low  $\rightarrow$  decrease  $Q$ ', ...
34     '\newline If  $\kappa$  is too high  $\rightarrow$  increase  $Q$ ', ...
35     '\newline Practical  $Q$  range: 600 to 3000'], ...
36     'Interpreter', 'latex', 'FontSize', 12, 'BackgroundColor', [0.9,0.9,0.9], ...
37     'EdgeColor', 'b', 'Position', [0.05,0.05,0.5,0.3]);

```

Listing 5: Practical guide for piano makers

3 Conclusion

The provided MATLAB codes allow the reader to simulate, measure, and control the quality factor in piano strings. These tools directly support the findings of the main paper: that Q is the dominant parameter governing the attack time, and that by adjusting Q (rather than geometry) one can precisely control the spectral friction parameter κ .

The complete set of codes and additional resources can be found in the GitHub repository:

<https://github.com/majidstar1234/spectralFriction-Revisited>